

13 C For geostationary satellite, $T = 24$ hours.

$$\text{For circular motion, } v = r\omega = \frac{2\pi r}{T} = \frac{2\pi(6.40 \times 10^6 + 3.59 \times 10^7)}{24 \times 60 \times 60} = 3.08 \text{ km s}^{-1}$$

14 B a t and x t graphs for a body undergoing s.h.m. are π rad out of phase